Socioeconomic and Demographic Drivers of Ambulatory Surgery Usage

The 2020 HCUP National Ambulatory Surgery Sample in 10 Charts

Scalable, Multi-Platform, Open-access, Reproducible Data Pipeline

7.8 million patients, 2899 hospitals, 35 states, 1 nation

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LLMs Utilized: Claude Sonnet 4 | Opus 4.1; ChatGPT 4o, o4, 5, 5-mini; Deepseek 3.1; Gemini 2.5 Pro; Grok 5

Riverside University HEALTH SYSTEM

AIM

To explore whether socioeconomics or demographics has more affect on the usage of ambulatory surgery.

METHODS

- Simple descriptive analysis → no statistical tests
- Vs 2020 US Census data → classical statistical tests
- Machine Learning → advanced statistical modeling
- Initially Traditional coding, later w/ use of AI/LLM

KEY FINDINGS

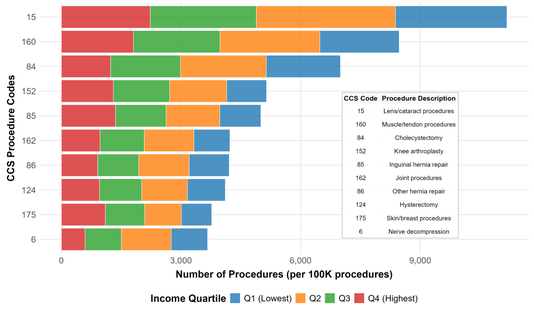
- Demographic shift over 1 American lifetime makes Age confounding
- Payor is the primary driver of ambulatory surgical access
- Medicare is the dominant payor, furthering Age confounding
- Geographical bias is a weakness of the dataset

CONCLUSION

- HCUP NASS data shows minimal socioeconomics imbalance
- Demographics generally reflect US population trends

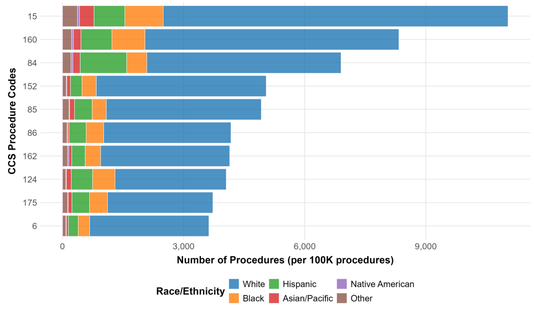
Most Common Ambulatory Surgeries by Socioeconomics

Segmented by Income Quartile of Patient ZIP Code



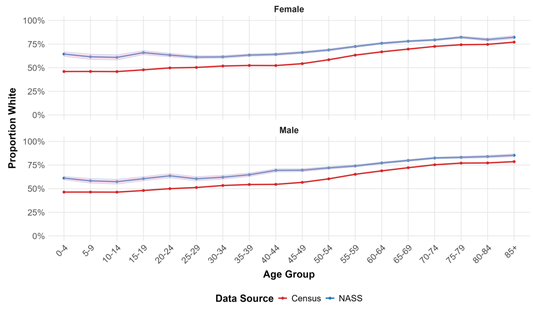
Most Common Ambulatory Surgeries by Demographic

Distribution by Patient Race/Ethnicity



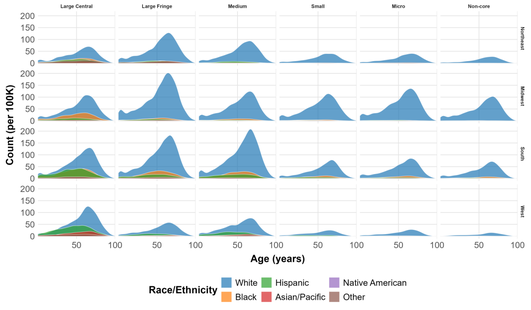
Race Proportion by Age Groups vs Census

Census comparions across age groups and gender (w/ CI)



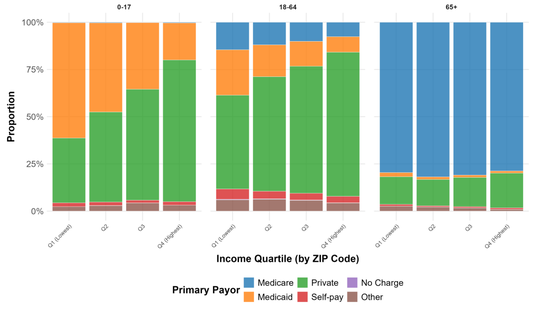
Patient Age Distribution in NASS 2020 by Demographic and Geography

By Urban-Rural Classification and US Region, Segmented by Race



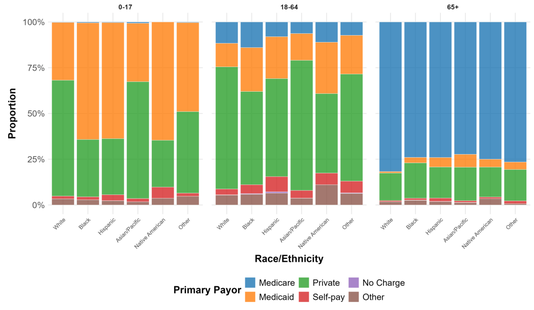
Primary Payor Distribution by Socioeconomics

Insurance by Age Group and Income Quartile



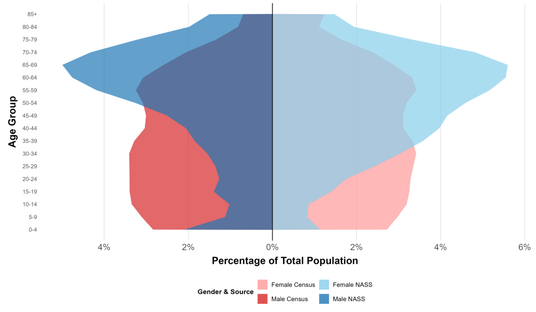
Primary Payor Distribution by Demographic

Insurance Coverage Patterns by Age Group and Race/Ethnicity



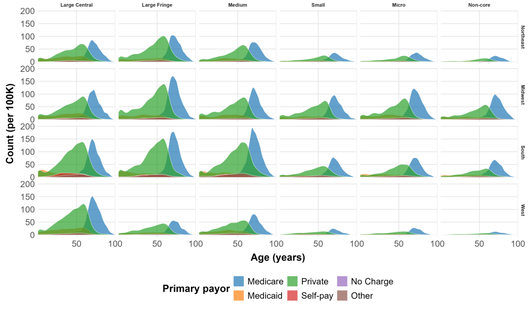
Demographic Proportions: NASS vs Census

Percentage of total population by 5-year age groups



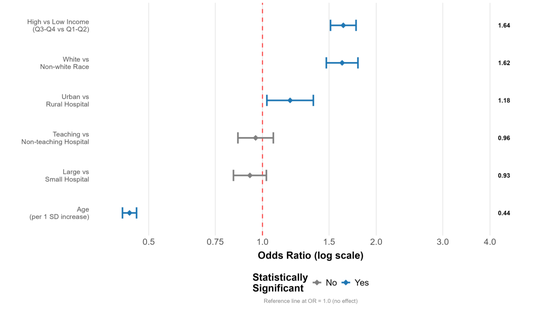
Patient Age Distribution in NASS 2020 by payor and Geography

By Urban-Rural Classification and US Region, Segmented by payor



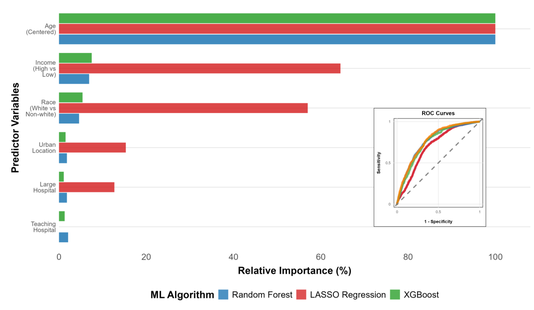
Predictors of Private Insurance

Multi-level Model Odds Ratios with 95% CI



Machine Learning Variable Importance

for Predicting Private Insurance Access; ROC Overlay



Run this analysis on your phone!

Generate this Poster

Explore Statistics and Code

Deploy your own VM

Scan here (Passcode: asa) →

Google Colab Link

| ASA 2025 | Data: HCUP NASS 2020 | Software: Python + R in Jupyterlab | Hardware: Local/Colab/VM |